Task-Dependent Neural Geometry Is Layer-Specific: An Exploratory Within-Subject Analysis of ds004511

Chart-Jacobian, one-step, and reachability estimates are distinct objects

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Abstract We asked whether resting state, cognitive control (CC), and gambling (GG) differ in explicitly non-interchangeable summaries of a subject-anchored neural chart. We reanalysed 134 EEG recordings from 45 OpenNeuro participants (45 Rest, 45 CC, 44 GG; one documented GG recording absent). Within-subject Wilcoxon tests and BH-FDR were applied in separate endpoint families. GG minus Rest increased the local chart-Jacobian spectral abscissa (median difference 0.01248, paired $d_z = 1.42$, $n = 44$, $q = 5.61 \times 10^{-}10$), whereas the independently identified lag-1 one-step spectral abscissa did not ($d_z = -0.33$, $n = 32$, $q = 0.18$). In the identified subset, GG reduced L2 rotation ($d_z = -1.10$, $q = 8.04 \times 10^{-}6$) and L3 generator rotation ($d_z = -1.08$, $q = 8.04 \times 10^{-}6$). Reachability amplitude summaries were available for 131/134 recordings; all three missing certificates failed the same numerical PSD gate, one in each task. The effective-dimension reachability readout was degenerate and is not interpreted. These export-level results establish a dissociation among estimators, not a neural mechanism, clinical biomarker, or confirmatory task effect.

Author Summary

The same EEG data can support different mathematical summaries of change. Gambling altered a local chart-Jacobian quantity but not its separately estimated one-step analogue. Occupancy, local flow, one-step dynamics, and reachability must therefore be tested and reported as different questions.

1. Introduction

Neural-state summaries are often discussed as though their names imply a common object. They do not. Occupancy describes where a recording lies in a declared chart; a local chart Jacobian describes a local derivative of that chart; an identified one-step map describes a fitted transition relation conditional on support; and a reachability summary describes a finite-horizon construction. Agreement between them is an empirical result, not an algebraic entitlement.

Core question

In a fixed, subject-anchored export, do task contrasts agree across local chart-Jacobian, identified one-step L2/L3, and reachability summaries, or do they expose estimator-specific structure and limitations?

2. Methods

2.1. Dataset and fixed measurement contract

The frozen export contains 134 recordings from 45 healthy participants: 45 Rest, 45 CC, and 44 GG recordings. One participant has no recorded GG EEG; this is retained as documented missingness rather than imputed. The input contract is the canonical MNPS `[m,d,e]` chart and its stratified 9D definition, with coordinates declared `subject_anchored`. No analysis retunes a measurement against task differences.

2.2. Endpoint families and statistics

The first family contains chart occupancy, local chart-Jacobian invariants, L1 local-law quantities, and recording-level reachability. The second, separate family contains chart-Jacobian alpha alongside lag-1 one-step L2/L3 quantities. Chart-Jacobian alpha is the spectral abscissa of a local chart Jacobian; L3 alpha is the spectral abscissa of the matrix-log generator proxy `log(Phi)/dt` for an identified direct lag-1 map (4 s), not the spectral

abscissa of `Phi` itself and not an Ito generator. They are not two labels for one estimator. Lag-2 was attempted, but only 13 direct L2 maps and 10 L3 proxies were computed (8 s; task-imbalanced), too sparse for a panel. L4 `reach_a_q` and `reach_v_norm` are amplitude summaries; rank-degenerate `reach_d_eff` is a QC finding, not a dimension contrast.

Recording values were reduced to subject-by-task medians. Pairwise later-minus-earlier contrasts used complete-case Wilcoxon signed-rank tests, paired d_z, and BH-FDR within the relevant panel. The L2/L3 pairwise panel's reported q is the global 36-test Wilcoxon family; its Friedman table is a separate 12-test family. Friedman tests use complete triplets. Bootstrap summaries use 10,000 draws with seed 20260921. This is an exploratory discovery analysis on one export: `exploratory_discovery_20260921_g4_not_locked`.

3. Results

3.1. R1: Coverage and quality control

All 134 recordings had a finite included chart-Jacobian alpha. Lag-1 one-step quantities were identified for 121/134 recordings (115/134 for the history-conditioned L2 quantity); hence the L2/L3 paired samples are smaller than the chart-Jacobian samples. Three recordings lacked finite included `reach_a_q`: one Rest, one CC, and one GG recording from different people. Each was marked invalid after `post_dtype_psd_failure` at gate F. This is a numerical certificate failure pattern, not evidence for task-specific absence.

Object	Finite recordings	Interpretive status
Chart-Jacobian alpha	134/134	Included; local-chart quantity
Lag-1 one-step L2/L3	121/134	Included only on identified support
History-conditioned L2	115/134	Secondary identified-support quantity
L4 amplitude summaries	131/134	Included with invalid certificates visible
L4 effective dimension	131/134	Method-limited; not tested as dimension

Table 1: Endpoint coverage. Missingness changes the eligible paired sample; it is not replaced by zeros.

3.2. R2: Chart-Jacobian and one-step alpha dissociate

GG minus Rest increased chart-Jacobian alpha (median difference 0.01248, $d_z = 1.42$, $n = 44$, $p = 1.56 \times 10^{-}11$, $q = 5.61 \times 10^{-}10$). The identified one-step L3 generator-proxy spectral abscissa did not show a positive or FDR-significant counterpart ($d_z = -0.328$, $n = 32$, $p = 0.129$, global-36-test $q = 0.178$). These are different complete-case subpopulations, not a formal test of a difference between operator effects; on the same 32 GG–Rest pairs, the chart-alpha contrast remains positive ($d_z = 1.461$, median difference 0.01271, $p = 3.26 \times 10^{-}9$). CC minus Rest likewise increased chart-Jacobian alpha ($d_z = 0.433$, $n = 45$, $q = 0.0167$), while the L3 proxy was uninformative on its smaller support ($d_z = -0.103$, $n = 32$, $q = 0.595$).

Primary dissociation

The GG–Rest chart-Jacobian alpha result is not a one-step L3 result. On separately defined and corrected complete-case samples, chart alpha is positive while the L3 generator proxy has no significant positive counterpart. This is not a formal between-estimator effect-difference test.

3.3. R3: Identified-support rotation results

On the 32 complete GG–Rest one-step pairs, GG decreased L2 `operator_rotation_rate` ($d_z = -1.10$, median difference −0.03589, $q = 8.04 \times 10^{-}6$) and L3 `one_step_generator_rotation_norm` ($d_z = -1.08$, median difference −0.05222, $q = 8.04 \times 10^{-}6$). These results are internally coherent within the identified subset, but they do not generalize automatically to the 13 recordings without lag-1 support.

GG minus Rest metric	n	d_z	p	BH q
Chart-Jacobian alpha	44	1.42	1.56e-11	5.61e-10
One-step L3 spectral abscissa	32	−0.328	0.129	0.178
L2 rotation rate	32	−1.10	6.91e-7	8.04e-6
L3 generator rotation norm	32	−1.08	6.91e-7	8.04e-6

Table 2: Key exploratory GG–Rest contrasts. Jacobian alpha and L2/L3 rows belong to distinct endpoint families.

3.4. R4: Reachability QC boundary

Finite L4 amplitude summaries differ strongly in the exploratory panel, but their interpretation is bounded by the certificate construction and the three invalid records. In contrast, 109 of 131 finite `reach_d_eff` values were within $10^{-}6$ of 1 (median 1; maximum 1.31377). Its small-residual Wilcoxon rows are therefore not reported as capacity or dimension results.

4. Discussion

4.1. What this study establishes

On this fixed export, task contrasts are layer-specific. A strong GG–Rest chart-Jacobian alpha contrast does not entail a positive, or even significant, one-step L3 alpha contrast. The identified subset instead shows reduced rotation summaries. This is a negative constraint on claims that one scalar can stand for local flow, an estimated one-step generator, or future accessibility.

4.2. Limitations and next gate

The analysis does not identify a biological mechanism of gambling, deception, or cognitive control. It does not establish causal control, individual-level prediction, a consciousness measure, or a validated mapping from MNPS to a latent neural operator. Subject anchoring, one export, partial L2/L3 support, and distinct BH families constrain the claims. A bounded QC audit found that all 13 absent lag-1 maps failed the existing out-of-fold fidelity gate (`one_step_fit_not_better_than_baseline`), despite regular 4-s grids and 120–260 lag-1 pairs; they comprised 12 Rest, 1 CC, and 0 GG recordings. This diagnoses a reproducible support boundary rather than restoring a map or resolving selection bias. Independent replication on a locked export and declared multiplicity plan is required before a confirmatory task claim.

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Bibliography

Author Contributions

RL: Conceptualization, Methodology, Software, Data Curation, Analysis, Writing, and Editing.

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Data Availability Statement

OpenNeuro ds004511 is publicly available under CC0. Article-local code, pinned configuration, result tables, and provenance are in the accompanying NeuralManifoldDynamics repository.

Competing Interests

The author developed NDT, MNPS, and software used in this analysis.